SESSION 18 – ELBOW METHOD + SILHOUETTE SCORE

🎯 Topic

- Problem of choosing K
- Elbow Method
- Inertia concept
- Silhouette Score
- Practical comparison

🎯 Class Objective

Students will:

- Understand why choosing K is important
- Learn **Elbow Method to find optimal K**
- Understand **Silhouette Score (quality of clusters)**
- Apply both methods practically

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1. 🧠 CLASS START (Hook)

Start like this:

"Kal humne K-Means me K=2 liya…
But kya wo best tha?
K=3 ya 4 better ho sakta hai?"

👉 Let students answer

Then say:

"Aaj hum seekhenge — correct K kaise choose karte hain."

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2. ❗ PROBLEM: HOW TO CHOOSE K?

Issue

- K manually set karna padta hai
- Wrong K → poor clustering

💡 Teaching Line

"Choosing K randomly is not correct."

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3. 📖 ELBOW METHOD

Concept

We run K-Means for multiple K values and observe error.

Metric Used

👉 **Inertia (WCSS – Within Cluster Sum of Squares)**

Meaning of Inertia

👉 Distance between:

- Data points
- Their cluster centroid

Goal

👉 Minimize inertia

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4. 📊 HOW ELBOW METHOD WORKS

Steps

1. Run K-Means for K = 1 to 10
2. Calculate inertia for each K
3. Plot graph
4. Find "elbow point"

Elbow Point

👉 Point where graph bends sharply

👉 Best K

💡 **Teaching Line**

"Elbow = where improvement suddenly slows down"

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5. 📈 VISUAL UNDERSTANDING

Explain:

- Before elbow → big improvement
- After elbow → small improvement

👉 So choose elbow point

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6. ⚠️ LIMITATION OF ELBOW METHOD

❌ Sometimes elbow is not clear
❌ Graph may be smooth

👉 So we use another method

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7. 📖 SILHOUETTE SCORE

Definition

Measures how well data points are clustered.

Range

Value	Meaning
+1	Good clustering
0	Overlapping clusters
-1	Wrong clustering

💡 **Teaching Line**

"Higher silhouette score = better clusters"

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8. 📊 INTUITION

Good clustering means:

- Points close to own cluster
- Far from other clusters

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9. 📊 ELBOW vs SILHOUETTE

Method	Purpose
Elbow	Find K using inertia
Silhouette	Check quality

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10. 🎯 INTERVIEW CONCEPTS

Q1: How to choose K?

👉 Elbow Method / Silhouette Score

Q2: What is inertia?

👉 Distance within cluster

Q3: What is silhouette score?

👉 Measure of cluster quality

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11. 💻 PRACTICAL (YOUR FORMAT)

🎯 Objective

Find best K using Elbow Method

13. 📊 OUTPUT INTERPRETATION

Elbow Graph

- Find bending point
- That K is best

Silhouette

- Highest score → best K

👉 Combine both for decision

In K-Means, choosing K is critical — and we use Elbow Method and Silhouette Score to make this decision scientifically.